Skills Summary

Fact Strategies: Turn It Around, Break It Apart

Use this reference while completing your worksheet or when studying.

Skill 1 — Turn-around facts (commutative)

Rule: The order of the factors never changes the product.
An array turned on its side still holds the same number of things.

Example: You know $4 \times 9 = 36$. Find 9×4.
Step 1. The factors are the same two numbers in the other order, so the product cannot change.
Step 2. $9 \times 4 = 36$, the same as 4×9.
$\Rightarrow$ **36**

Try it: You know $3 \times 8 = 24$. What is 8×3?

check: **24**

Skill 2 — Break-apart facts (distributive)

Rule: Split one factor into easy pieces, multiply each piece, add the parts.
Good splits use your anchor facts: fives, tens, and twos.

Example: Find 7×8 by breaking the 8 apart.
Step 1. Split the 8 into $5 + 3$, two pieces with friendly facts.
Step 2. $7 \times 5 = 35$ and $7 \times 3 = 21$, so $35 + 21 = 56$.
$\Rightarrow$ **56**

Try it: Find 6×12 by splitting the 12 into $10 + 2$.

check: **72**

Skill 3 — Choosing the strategy

Rule: Ask two questions, in order:
Do I know the turn-around fact already? If yes, use it. If not, which factor splits into fives, tens, or twos most easily? Break that one.

Example: Choose a strategy for 8×6.
Step 1. Neither order is a fact I know cold, so break a factor: the 6 splits into $5 + 1$, both easy with 8.
Step 2. $8 \times 5 = 40$ and $8 \times 1 = 8$, so $40 + 8 = 48$.
$\Rightarrow$ **48**

Try it: Choose your own strategy for 7×9 and solve it.

check: **63**

(!) Watch out: When you break a factor apart, multiply *both* pieces — then *add* the two part-products. Forgetting one piece (or adding the pieces before multiplying) is the classic slip.